

# development\_history

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## MIDAS GUI — Development History

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A commit-by-commit record of what changed and when, so you can (a) find **when** a particular behaviour/feature entered the code and (b) know **what to roll back** to undo a specific effect.

Keep this file updated when you add commits: append a new entry at the bottom of the chronological log and add a row to the quick-reference index. Regenerate the PDF the same way as the other docs (pandoc + headless Chrome) if you keep one.

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## How to use this document

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- **Find when a change landed:** search the change → commit index below, or `git log --oneline -- <path>` for a specific file, or `git log -S"<code string>"` to find the commit that introduced/removed a string.
- **See a commit's exact diff:** `git show <hash>` (whole commit) or `git show <hash> --<path>` (one file).
- **Undo one commit cleanly (keeps later history):** `git revert <hash>`. For a range: `git revert <oldest>^..<newest>`.
- **Undo just one file back to a commit:** `git checkout <hash> -- <path>` then commit. To see the file *as of* a commit without changing anything: `git show <hash>:<path>`.
- **Hard reset to a point (destructive — local only, never on pushed history):** `git reset --hard <hash>`. Prefer `git revert` on a shared branch.
- **Dependencies matter.** Many later commits build on earlier ones (e.g. the unified data-loader, the config overlay). The "Roll back" note on each entry flags when reverting will disturb dependents.

Branch: `main`. All commits are authored by Dishant Beniwal (AI pair-authored). Dates are commit dates (YYYY-MM-DD).

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## Change → commit quick-reference index

### Packaging / build / launch

| Change                                                                    | Commit                  |
|---------------------------------------------------------------------------|-------------------------|
| Initial v1.0.0 release (9-tab app, all backends)                          | <a href="#">f19ee8c</a> |
| setuptools.build_meta backend (broader pip compat)                        | <a href="#">3cd1f36</a> |
| <code>launch.py</code> standalone launcher                                | <a href="#">2e9f419</a> |
| Startup crash diagnostics + robust launcher (Windows)                     | <a href="#">916ece2</a> |
| MIDAS-style packaging (LICENSE, pyproject, release.sh, smoke tests)       | <a href="#">e862135</a> |
| Ship test_data sample data in the repo (fresh clone works out of the box) | <a href="#">7e157e0</a> |
| Drop midas_suite from env (unblock conda env create; backends via -e .)   | <a href="#">72a1770</a> |
| Pin environment.yml to verified NumPy-1 set + README recreate steps       | <a href="#">68313f8</a> |

### Data Viewer (Tab 0)

| Change                                                                          | Commit                  |
|---------------------------------------------------------------------------------|-------------------------|
| Radial-integration plot, beam-centre ring picking, zoom persistence             | <a href="#">acf0866</a> |
| Zoom/pan preserved across frames (autoRange off)                                | <a href="#">d9e527e</a> |
| Intensity-range mask, calibration-file loading, click-to-draw ring              | <a href="#">d50b588</a> |
| Compact 3-per-row lattice + cubic quick-set; mask default max(99.99pct, 1e5)    | <a href="#">41f28f5</a> |
| Projection Skip-frames field; compact geometry/intensity fields; Lsd separators | <a href="#">108ea7d</a> |
| Intensity-statistics panel + histogram (bottom of loader)                       | <a href="#">2b7a695</a> |
| Tilt/distortion-aware radial integration when a calibration is loaded           | <a href="#">df544d2</a> |
| Data Viewer Top-N brightest pixels + mask-aware stats + full-geometry receive   | <a href="#">bc86d74</a> |
| Calibrate per-coefficient distortion, frame averaging, seed feedback            | <a href="#">b0a5cc3</a> |
| Batch read-only "Calibration values" card                                       | <a href="#">b6f1681</a> |
| Pump Probe publication plotting, delay colour-bar, default detector mask        | <a href="#">49623ae</a> |
| Gitignore local context, internal docs, large sample sets                       | <a href="#">b251ca8</a> |
| Live Data card: in-tab EPICS PVA stream via pvapy, reload button                | <a href="#">1769f69</a> |
| Live viewer: manual colormap/level changes now survive new frames               | <a href="#">234ded9</a> |

| Change                                                                 | Commit  |
|------------------------------------------------------------------------|---------|
| Wider image/radial-plot splitter handle                                | 69f470c |
| Radial profile: Y-min defaults to 0.9x data min, manual Y-range sticks | 96f8add |
| Simulate rings is now a live toggle (auto-recomputes on param change)  | ecf6d01 |
| Live PV field becomes a device dropdown (Preferences ▶ Devices)        | aa82cb6 |
| Fix image-view autorange drift; bound pan/zoom to image                | c156c62 |

## Mask Builder (Tab 1)

| Change                                                                                                | Commit  |
|-------------------------------------------------------------------------------------------------------|---------|
| Unbounded $\sigma$ fields; independent spatial/temporal auto-mask; Viewer↔Calibrate geometry hand-off | 10df828 |
| Temporal-constancy accepts a 3-D HDF5 (time,y,x) stack                                                | 2385f75 |

## Calibrate (Tab 2) / Refinement (Tab 3)

| Change                                                                            | Commit  |
|-----------------------------------------------------------------------------------|---------|
| Abort buttons; dark/bright/background field correction; auto-load on browse       | 7d010d7 |
| Calibration input accepts paramtest/poni/json                                     | 41f28f5 |
| Load-calibration-file (geometry + $\lambda$ ) helpers                             | b381d8d |
| Multi-column paramtest results readout + Send-to-Data-Viewer button               | 455ca4e |
| Calibrate Results all-text (drop distortion table; named distortion, bigger font) | 3a6ecb9 |
| Fix calibrate() initial_BC_y TypeError (kwargs filter) + pin scikit-image         | b1642fa |
| Pick-Ring fit overlay recolored blue (was same amber as simulated rings)          | 3eb4a44 |

## Batch Integrate (Tab 4)

| Change                                                                       | Commit  |
|------------------------------------------------------------------------------|---------|
| Live folder MONITOR + per-file legend                                        | 1149afc |
| Clear results + inline per-file labels & plot controls                       | 524f5f1 |
| Stacked profiles: publication themes, point+line, symbol size, x-units, grid | d135c80 |
| Waterfall R/2 $\theta$ /Q x-axis selector                                    | 2385f75 |

## PDF Analysis (Tab 6)

| Change                                                                             | Commit  |
|------------------------------------------------------------------------------------|---------|
| Polyatomic midas_pdf reduction pipeline + vendored midas_pdf + calibration loading | b381d8d |
| Defaults to real I(Q) test data                                                    | 1149afc |

## Pump Probe (TR-XRD) tab

| Change                                                                                                                                                    | Commit  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| New tab: TRR pooling + MIDAS-engine integration + $\Delta I(q, \text{delay})$ + 4 pyqtgraph views + publication-quality plots + default MIDAS calibration | 590b410 |

## Cross-cutting UI / infrastructure

| Change                                                                        | Commit  |
|-------------------------------------------------------------------------------|---------|
| Linux visible files/folders style fix                                         | fc754d0 |
| Pixel-weighted azimuthal mean; readable QMenu; no-scroll combos               | 41f28f5 |
| Unified Data-Loader panel + draggable 3-panel splitter (tabs 0/2/3/4)         | aa9395e |
| Clickable $\lambda$ label with K-edge foil menu                               | b628446 |
| Clickable px label with detector pixel-size menu                              | 3248638 |
| Per-user JSON configuration system + Preferences dialog                       | d30be2c |
| Modular tab visibility (show/hide optional tabs)                              | 590b410 |
| User-adjustable interface scaling (HiDPI / 4K, QT_SCALE_FACTOR)               | d5fafd8 |
| Default optional tabs reduced (Corrections/PDF/Texture/Export hidden)         | 7e157e0 |
| Lsd displayed in mm (calculations & calibration files stay in $\mu\text{m}$ ) | c4e1c12 |
| Fix "Populating font family aliases" warning (real fixed-width fonts)         | d6df1f8 |
| Fix fresh-env launch crash — colormap resolution without matplotlib           | 6332b3d |
| Fix native Bus-error crash on Run Calibration — cap BLAS/OMP thread pools     | 0b4326d |
| Preferences ▶ Devices tab; Live PV field becomes a device dropdown            | aa82cb6 |

## Stability / performance / consistency (review-driven, phases 1–3)

| Change                                                                                   | Commit  |
|------------------------------------------------------------------------------------------|---------|
| Clean worker shutdown, drop QThread.terminate(), guard stdout, offload projection        | 399f3d7 |
| Waterfall $O(N^2) \rightarrow O(N)$ , frame-slider debounce, radial-grid + stats caching | 4462ee1 |
| Dedupe distortion map + paramstest writer, align calibrant lattices, texture colormap    | 8846619 |

## Documentation

| Change                                                              | Commit  |
|---------------------------------------------------------------------|---------|
| README not-on-PyPI clone/conda instructions                         | cd05ced |
| gui_documentation.md added                                          | 2e5f7c9 |
| gui_documentation.pdf added                                         | 75c135c |
| README/environment install via <code>pip install midas_suite</code> | aceb40f |
| implementation_details.md/.pdf added                                | 524f5f1 |
| config_gui.md + config.example.json added                           | d30be2c |
| development_history.md/.pdf added                                   | 6d6f3fb |

## Chronological commit log (oldest → newest)

### f19ee8c — Initial release: midas\_gui v1.0.0 (2026-06-30)

**Effect:** First version. Nine-tab PyQt5 GUI over `midas_calibrate_v2` / `midas_integrate_v2`: mask building, auto-calibration, refinement, batch integration + waterfall, corrections, PDF, texture, export. Entry points `midas-gui` / `python -m midas_gui`. **Files:** entire `midas_gui/` package + `pyproject.toml`, `README`, `environment.yml`. **Roll back:** this is the root — everything depends on it; don't revert.

### 3cd1f36 — setuptools.build\_meta backend (2026-06-30)

**Effect:** Build backend switched for `setuptools≥61` compatibility. **Files:** `pyproject.toml`. **Roll back:** `git revert 3cd1f36` (only affects builds).

### 2e9f419 — add launch.py standalone launcher (2026-06-30)

**Effect:** `python /path/to/launch.py` runs the app from anywhere. **Files:** `launch.py`. **Roll back:** safe to revert; later hardened by `916ece2`.

### `fc754d0` — Linux style fix for invisible files/folders (2026-06-30)

**Effect:** Stylesheet fix so file/folder text is visible on Linux. **Files:** `midas_gui/style.py`. **Roll back:** `git revert fc754d0` (cosmetic).

### `acf0866` — Data Viewer: radial plot, ring picking, zoom persistence (2026-06-30)

**Effect:** Added the radial-integration plot, beam-centre ring picking, and zoom-persistence across a stack in the Data Viewer; widened left panels. **Files:** `tab_view.py` (+1-line width bumps in other tabs). **Roll back:** revert to remove the Data Viewer radial plot/ring picking.

### `d9e527e` — preserve zoom/pan across frames (2026-06-30)

**Effect:** Disabled `pyqtgraph autoRange` on redisplay so zoom/pan survive frame changes. **Files:** `widgets.py`. **Roll back:** revert if you *want* auto-refit per frame.

### `d50b588` — Data Viewer: intensity mask, calib loading, click-to-draw ring (2026-07-01)

**Effect:** 99.9-pct intensity-range mask default; load calibration (`json/poni/paramstest`); click-to-draw ring; test-data defaults point at local `test_data/`. **Files:** `tab_view.py`, `widgets.py`, `helpers.py`, `constants.py`, `tab_mask.py`. **Roll back:** revert to drop these Data Viewer features (note `helpers.py` geometry parsing added here is reused later — see `b381d8d`).

### `916ece2` — startup crash diagnostics + robust launcher (2026-07-01)

**Effect:** Logs uncaught exceptions/native faults to `~/midas_gui_error.log`, prevents PyQt slot-exception aborts, isolates failing tabs, hardens the launcher. **Files:** `app.py`, `launch.py`. **Roll back:** revert only if you want raw exceptions to propagate; not recommended.

### `7d010d7` — Abort buttons + dark/bright/background + auto-load (2026-07-01)

**Effect:** Abort on calibrate/batch; dark/bright/background field correction (file/folder/HDF5, index-range averaging, divide/subtract); browse auto-loads, "Load" buttons removed; compact FieldSelector. **Files:** `tab_batch/calibrate/corrections/mask/pdf/refine/texture/view.py`, `widgets.py`, `workers.py`, `helpers.py`. **Roll back:** wide-reaching; reverting removes field

correction + auto-load across tabs. The Abort behaviour here was later re-worked by 399f3d7 (no `terminate()`).

### e862135 — MIDAS-style packaging (2026-07-01)

**Effect:** BSD-3 LICENSE, MIDAS-convention `pyproject.toml`, `release.sh` + `RELEASING.md`, headless `tests/` smoke suite. **Files:** `LICENSE`, `README.md`, `RELEASING.md`, `pyproject.toml`, `release.sh`, `tests/`. **Roll back:** revert affects packaging/tests only.

### 41f28f5 — compact lattice, pixel-weighted azimuthal mean, readable menus (2026-07-06)

**Effect:** Data Viewer 3-per-row lattice + cubic quick-set; intensity-mask default `max(99.99pct, 1e5)`; **pixel-weighted azimuthal mean** (selectable vs legacy  $\eta$ -bin mean) fixing off-detector-BC distortion; batch calibration accepts `paramtest/poni/` json; readable right-click menus; no-scroll combos. **Files:** `helpers.py`, `workers.py`, `tab_view/batch/calibrate/pdf/refine/texture.py`, `widgets.py`, `style.py`. **Roll back:** to undo the azimuthal-averaging change specifically, prefer editing the `weighted` default rather than reverting the whole commit (it bundles UI changes).

### cd05ced — docs: not-on-PyPI install instructions (2026-07-06)

**Effect:** README clone+conda+run instructions; relative env self-install. **Files:** `README.md`, `environment.yml`. **Roll back:** docs only.

### 2e5f7c9 — docs: add gui\_documentation.md (2026-07-06)

**Effect:** First committed user guide. **Files:** `documentation/gui_documentation.md`. **Roll back:** docs only.

### 75c135c — docs: add gui\_documentation.pdf (2026-07-06)

**Effect:** PDF build of the guide. **Files:** `documentation/gui_documentation.pdf`. **Roll back:** docs only.

### aceb40f — docs: install via `pip install midas_suite` (2026-07-06)

**Effect:** Corrected backend-install instructions. **Files:** `README.md`, `environment.yml`. **Roll back:** docs only.

## aa9395e — Unified Data-Loader panel + 3-panel splitter (tabs 0/2/3/4) (2026-07-07)

**Effect:** Shared `DataLoaderPanel` (Data/Dark/Bright/Background/Mask, per-mode frame controls) + `MaskSelector`; corrections applied on tabs 0/2/3/4; each of those tabs restructured into a draggable `[Data Loader | Parameters | Display]` splitter. **Files:** `widgets.py` (+442), `tab_view/calibrate/batch/refine.py` (large rewrites), `app.py`, `helpers.py`, `workers.py`, docs. **Roll back: high impact** — this is the base for the Batch/Calibrate/Refine and later Pump Probe layouts. Reverting breaks the 3-panel tabs and the Pump Probe tab's left panel. Avoid a blanket revert; fix forward instead.

## b381d8d — Tab 6 polyatomic midas\_pdf pipeline + calibration loading (2026-07-08)

**Effect:** PDF tab rewritten to the total-scattering pipeline; **vendored** `midas_pdf` under `midas_gui/_vendor/` + `pdf_backend.py`; helpers `geometry_fields_from_file` / `result_ns_from_geometry_file` (paramtest/json/poni) used by Calibrate and PDF "Load calibration file...". **Files:** `_vendor/midas_pdf/**` (new, large), `pdf_backend.py`, `tab_pdf.py`, `workers.py`, `helpers.py`, `tab_calibrate.py`, `constants.py`, `pyproject.toml`, docs. **Roll back:** reverting removes the PDF pipeline **and** the shared geometry-file loaders that later tabs (incl. Pump Probe via `spec_from_geometry_file`) rely on — revert with care.

## 1149afc — Batch live MONITOR + legend; PDF real-data default (2026-07-08)

**Effect:** MONITOR button (stream mode) watches a folder and integrates new frames via `FolderMonitorWorker`, reusing a cached detector map (factored `build_integration_context()` + `write_profile()`, shared with `BatchWorker`); per-file legend; PDF tab defaults to real I(Q). **Files:** `tab_batch.py`, `workers.py`, `widgets.py`, `constants.py`, `tab_pdf.py`, docs. **Roll back:** to remove MONITOR only, revert this commit — but `build_integration_context()` introduced here is **reused by the Pump Probe worker** ( 590b410 ), so a full revert would break Pump Probe integration.

## 524f5f1 — Batch Clear results + inline labels; implementation-details doc (2026-07-09)

**Effect:** "Clear results" button; inline per-file curve labels + line-width/font toolbar on stacked profiles; new `implementation_details.md/.pdf`. **Files:** `tab_batch.py`, `widgets.py`, `documentation/implementation_details.*`. **Roll back:** safe, localized to Batch + the new doc.



## **10df828 — Mask unbounded $\sigma$ + split auto-mask; Viewer↔Calibrate geometry (2026-07-09)**

**Effect:** Removed  $\sigma$  upper limits; split statistical auto-mask into independent Spatial-outlier / Temporal-constancy checkboxes; geometry hand-off `DataViewer.pushGeometry / Calibrate.pullGeometry` → `apply_geometry`. **Files:** `tab_mask.py`, `tab_view.py`, `tab_calibrate.py`, `app.py`, `workers.py`, docs. **Roll back:** revert to restore bounded  $\sigma$  / combined auto-mask / no geometry hand-off.

## **d135c80 — Batch stacked-profiles: themes, point+line, x-units, grid (2026-07-09)**

**Effect:** `StackedProfileViewer` publication/dark themes, point+line, symbol-size, R/2 $\theta$ /Q x-unit selector, grid toggle. (This viewer's styling is the template the Pump Probe plots later mirror.) **Files:** `widgets.py` (+228), `tab_batch.py`, docs. **Roll back:** localized to the Batch stacked-profile viewer.

## **2385f75 — Batch waterfall R/2 $\theta$ /Q axis; Mask 3-D HDF5 stack (2026-07-10)**

**Effect:** Waterfall x-unit selector via a converting `AxisItem + _convert_radial()`; Mask temporal-constancy reads a single 3-D (time,y,x) HDF5. **Files:** `widgets.py`, `tab_batch.py`, `tab_mask.py`, `workers.py`, docs. **Roll back:** `_convert_radial / _UnitAxis` introduced here are reused by the Pump Probe heatmap — don't fully revert if Pump Probe is present.

## **b628446 — clickable $\lambda$ label with K-edge foil menu (2026-07-10)**

**Effect:** Compact clickable " $\lambda$ " label → K-edge foils menu (sets  $\lambda$  from edge energy). `constants.K_EDGE_FOILS`, `HC_KEV_A`, `helpers.make_kedge_label`. **Files:** `constants.py`, `helpers.py`, `tab_view/calibrate/pdf.py`, docs. **Roll back:** localized; `make_pixel_label` (next) reuses the factored helper.

## **108ea7d — Data Viewer projection Skip-frames + compact fields (2026-07-10)**

**Effect:** Projection "Skip frames" field; narrower Axis/Skip/pixel fields; Lsd thousands-separators. **Files:** `tab_view.py`, docs. **Roll back:** localized.

## **3248638 — clickable px label with detector pixel-size menu (2026-07-10)**

**Effect:** Clickable "px" label → GE/Varex/Pilatus/Eiger sizes; `constants.PIXEL_PRESETS`; factored `helpers._clickable_menu_label`. **Files:** `constants.py`, `helpers.py`, `tab_view.py`, `tab_calibrate.py`, docs. **Roll back:** localized.

## 2b7a695 — Data Viewer intensity-statistics panel + histogram (2026-07-10)

**Effect:** `IntensityStatsPanel` (histogram + p70/p90/p99/p99.9/p99.99 with pixel counts) pinned to the loader bottom (stack mode); Current/All/projected scope. **Files:** `widgets.py`, `tab_view.py`, `docs`. **Roll back:** localized.

## d30be2c — per-user configuration system + Preferences dialog (2026-07-12)

**Effect:** Per-user JSON config overlays shipped defaults at import; `settings.py`; `prefs_dialog.py` (Geometry/Paths/Materials/Calibrants/Menus/Algorithms); Settings menu; `DEFAULT_KERNEL/PIPELINE/OUTPUT_FORMAT/ERROR_MODEL/COLORMAP`; tab combos honor configured defaults. New `config_gui.md`, `config.example.json`, `tests/test_config.py`. **Files:** `settings.py` (new), `prefs_dialog.py` (new), `constants.py` (+165), `app.py`, `tab_batch.py`, `tab_calibrate.py`, `widgets.py`, `docs`, `tests`. **Roll back: high impact** — the config overlay + `DEFAULT_*` and the Preferences dialog are extended by the modular-tabs commit ( 590b410 ). Reverting this would also break `ui.visible_tabs` and the Tabs preferences section.

## 399f3d7 — Stability: clean worker shutdown, no terminate(), stdout guard (2026-07-12)

**Effect:** `closeEvent` stops all tab QThreads; removed every `QThread.terminate()` (cooperative interrupt + orphan instead); `CalibrationWorker` restores stdout only if still its own; Data Viewer projection moved to `ProjectionWorker` (off GUI thread). **Files:** `app.py`, `tab_batch.py`, `tab_calibrate.py`, `tab_view.py`, `workers.py`. **Roll back:** reverting reintroduces the exit hard-crash risk and UI-freeze on projection — not recommended. Phase 1 of the 3-part review.

## 4462ee1 — Performance: waterfall O(N), debounce, caching (2026-07-12)

**Effect:** Waterfall pre-allocated buffer + 100 ms coalesced redraw ( $O(N)$  not  $O(N^2)$ ); 60 ms frame-slider debounce; cached radial bin-index grid; skip all-frame stats on frame-only changes. **Files:** `tab_view.py`, `widgets.py`. **Roll back:** reverting slows large scans / scrubbing; behaviour otherwise same. Phase 2 of the review.

## 8846619 — Consistency: dedupe maps/writer, align lattices (2026-07-12)

**Effect:** `helpers._PARAMSTEST_DISTORTION` derived from `constants._V2_T0_V1` (removed duplicated `_V2V1` dicts); shared `helpers.write_standalone_paramstest` used by Calibrate + Export; LaB6/Si `_LATT/_LC` aligned to `MATERIALS`; texture colormap honors `DEFAULT_COLORMAP`.

**Files:** `constants.py`, `helpers.py`, `tab_calibrate.py`, `tab_export.py`, `tab_texture.py`. **Roll back:** reverting reintroduces duplicated code paths; low functional risk. Phase 3 of the review.

## 590b410 — Modular tabs + Pump Probe (TR-XRD) tab + plot quality (2026-07-12)

**Effect:** (1) **Modular tab visibility** — Data Viewer/Mask/Calibrate/Batch always shown, the rest toggled in Settings ▶ Preferences ▶ Tabs (`ui.visible_tabs`, applied live); `app.apply_tab_visibility()`. (2) **New Pump Probe tab** (`tab_pumpprobe.py`) — TRR filename pooling, MIDAS-engine integration via new `PumpProbeWorker` (reuses `build_integration_context / integrate_frame`),  $\Delta I(q, \text{delay})$ , three-panel layout, four pyqtgraph views; ships a converted MIDAS `paramstest` as the default calibration. (3) **Plot quality** — white publication theme, black axes/labels/titles, readable legends,  $\Delta I$  colorbar, aligned reference panel, shared draw-mode + line/point/font-size toolbar, bounded pan/zoom. **Files:** `tab_pumpprobe.py` (new), `workers.py` (+ `PumpProbeWorker`), `app.py`, `constants.py`, `prefs_dialog.py`, `tests/test_smoke.py`, `docs` (`gui_documentation`, `config_gui`, `config.example.json`). **Roll back:** to remove **only Pump Probe**, delete `tab_pumpprobe.py` and its wiring in `app.py / workers.py`; to remove **only modular tabs**, revert the `app.apply_tab_visibility / prefs_dialog` Tabs section and `constants` tab lists. A full `git revert 590b410` removes both features together. Depends on `aa9395e` (`DataLoaderPanel`), `1149afc` (`build_integration_context`), `2385f75` (`_convert_radial / _UnitAxis`), `d30be2c` (`config/prefs`).

## 6d6f3fb — docs: add development\_history.md/.pdf (2026-07-13)

**Effect:** Added this per-commit change log + change→commit index + rollback guide. **Files:** `documentation/development_history.md`, `documentation/development_history.pdf`. **Roll back:** docs only.

## d5fafd8 — UI scaling: user-adjustable whole-interface zoom (HiDPI / 4K) (2026-07-13)

**Effect:** Whole-application scale (layout + fonts) for HiDPI / 4K monitors. `constants.DEFAULT_UI_SCALE` (`config ui.ui_scale`); `app.main()` applies it via `QT_SCALE_FACTOR` **before** the `QApplication` is created (so fixed widths, splitters, pyqtgraph and stylesheet px all scale uniformly) + `AA_UseHighDpiPixmaps`. Manual control in **Preferences ▶ Display** (spin + 100/125/150/200 % presets) and **Settings ▶ Interface scaling...**; both persist `ui.ui_scale` and offer an immediate self-restart (`MainWindow.restart_app` via `QProcess`) since the factor is read only at startup. **Files:** `constants.py`, `app.py`, `prefs_dialog.py`, `docs` (`gui_documentation`, `config_gui`, `config.example.json`). **Roll back:** `git revert d5fafd8`. Self-contained (depends only on the config system `d30be2c`); reverting removes the scale control and the app renders at 1.0x. To keep the feature but disable it, set `ui.ui_scale` to 1.0.

## c4e1c12 — Display Lsd in mm (calculations & files stay in $\mu\text{m}$ ) (2026-07-13)

**Effect:** The sample-to-detector distance is entered/shown in **mm** in the Data Viewer, the Calibrate seed, and Preferences ▶ Geometry; conversion to/from  $\mu\text{m}$  happens only at the display boundary ( `DataViewerTab._lsd_um()` ,  $\times 1000$  on read,  $\div 1000$  on set).

`get_geometry` / `apply_geometry` / `manual_seed` / `prefs` `_assemble` still emit  $\mu\text{m}$ ; the config key stays `lsd_um` ( $\mu\text{m}$ ); calibration-file writers ( `paramtest.txt` , `calibration.json` ) are unchanged (always  $\mu\text{m}$ ). Batch drift-status label → mm. **Files:** `tab_view.py` , `tab_calibrate.py` , `prefs_dialog.py` , `tab_batch.py` , docs. **Roll back:** `git revert c4e1c12` to return every Lsd field to a  $\mu\text{m}$  display. Purely a display-layer change — no stored/file/calculation values are affected either way.

## d6df1f8 — Fix "Populating font family aliases" startup warning (2026-07-13)

**Effect:** Removed the `qt.qpa.fonts: Populating font family aliases` warning (and its ~40 ms startup cost) that Qt emitted because the code named the nonexistent family "Monospace" — both via `QFont("Monospace", ...)` and the CSS generic `font-family:monospace` . Now names real per-platform families (Menlo/Consolas/DejaVu Sans Mono/Courier New) via `style.MONO_FAMILIES` / `MONO_CSS` and `widgets._mono_font()` (`QFont` `setFamilies` + Monospace style hint). **Files:** `style.py` , `widgets.py` , `tab_export.py` , `tab_view.py` . **Roll back:** `git revert d6df1f8` (cosmetic; only affects fixed-width font selection and re-introduces the harmless warning).

## 455ca4e — Calibrate: multi-column results readout + Send-to-Data-Viewer (2026-07-13)

**Effect:** (1) The Calibrate **Results** tab shows the full parameter set exactly as written to `paramtest.txt` (Lsd, BC, tx/ty/tz, p0–p14, Parallax, Wavelength, px, NrPixelsY/Z, RhoD, SpaceGroup, LatticeConstant) in a 3-column, column-major grid ( `_paramtest_pairs` via the shared writer + `_populate_param_grid` ), with a strain/timing line and the named-distortion table below. (2) New → **Send to Data Viewer** button ( `sendGeometryToViewer` signal + `_send_to_viewer` ) pushes the calibrated  $\lambda/\text{pixel}$ /Lsd/beam-centre into the Data Viewer through new `DataViewerTab.set_geometry` ( $\mu\text{m}$  internal, Lsd shown mm, auto-BC off); wired in `app.py` as the reverse of the Viewer→Calibrate hand-off. **Files:** `tab_calibrate.py` , `tab_view.py` , `app.py` , docs. **Roll back:** `git revert 455ca4e` . Self-contained; depends on the mm-display change ( `c4e1c12` ) for the Viewer Lsd conversion.

## 7e157e0 — Ship test\_data; hide four optional tabs by default (2026-07-14)

**Effect:** (1) `test_data/` sample data (`calibrant_ceria.tif/h5`, `nickel_stack.h5`, `nickel_tifs/`, `make_test_data.py`) is now committed so the GUI's default paths work on a fresh clone — `.gitignore` re-includes it past the global `*.tif / *.h5` ignores via trailing negations, while `test_data/output/`, `out_temp.txt` and `.DS_Store` stay ignored. (2) `DEFAULT_VISIBLE_TABS = ["Calib. Refinement", "Pump Probe"]` — Corrections, PDF Analysis, Texture and Results & Export ship hidden (enable in Preferences ▶ Tabs). **Files:** `.gitignore`, `test_data/**` (added), `constants.py`, `config.example.json`, `tests/test_smoke.py`, `docs`. **Roll back:** `git revert 7e157e0` restores the all-optional-tabs default and removes the shipped data + re-ignores `test_data/`. Note: a per-user config with its own `ui.visible_tabs` overrides the shipped default regardless.

## 6332b3d — Fix launch crash on fresh Linux/Windows (colormap w/o matplotlib) (2026-07-14)

**Effect:** Fixes the GUI failing to start on a clean env where every always-on tab's image viewer crashed with `'NoneType' object has no attribute 'getColors'`. Cause: `DEFAULT_COLORMAP` "hot" (and the whole `COLORMAPS` list) are matplotlib colormaps; `pg.colormap.get("hot")` returns `None` without matplotlib, and that `None` was passed into `pyqtgraph`. New `widgets._resolve_cmap()` resolves native → matplotlib → native fallback → grayscale ramp and never returns `None` (used by `ImageViewer`, `WaterfallViewer`, `Texture`); matplotlib added as a declared dependency (`pyproject` + `environment.yml` `matplotlib-base`) so the named maps actually resolve. The downstream "Geometry hand-off wiring failed / QWidget has no attribute `pushGeometry`" log lines were only a symptom of the tabs becoming placeholders and disappear with this fix. **Files:** `widgets.py`, `tab_texture.py`, `pyproject.toml`, `environment.yml`, `tests/test_smoke.py`. **Roll back:** `git revert 6332b3d` (reintroduces the crash on matplotlib-less envs).

## 72a1770 — env: drop midas\_suite (unblocks `conda env create`) (2026-07-14)

**Effect:** `conda env create -f environment.yml` was aborting because the pip section's `midas_suite` meta-package pulls `midas-index 0.7.3`, whose `sdist` fails to build against `scikit-build-core>=0.8` (Use `cmake.version` instead of `cmake.minimum-version`). The GUI never uses `midas_suite`'s extras; the backends it actually imports (`midas-calibrate-v2/-integrate-v2/-calibrate/-hkl/-distortion`) are declared in `pyproject.toml` and pulled by the editable `-e .` install. Removed the redundant `midas_suite` line; README updated. (Unrelated to the repo: the `libarchive.19.dylib` / `conda-libmamba-solver` errors in that log are a broken base-Anaconda mamba, worked around with `--solver=classic`.) **Files:** `environment.yml`, `README.md`. **Roll back:** `git revert 72a1770` (re-adds `midas_suite` and its build failure).

## 68313f8 — Pin environment.yml to a verified NumPy-1 set + README recreate steps (2026-07-14)

**Effect:** Makes `conda env create` reproduce a known-good environment. Fresh installs previously pulled the newest backends (numba 0.66 → NumPy 2.x) against conda torch 2.2.2, which can't interop with NumPy 2 (`torch.from_numpy: Numpy is not available`). Pinned every package to the versions from the user's working base Anaconda — the **NumPy-1 family**: `numpy=1.26.4`, `scipy=1.13.1`, `h5py=3.11.0`, `tiffle=2023.4.12`, `pyqtgraph=0.14.0`, `matplotlib-base=3.8.4` (conda); `PyQt5==5.15.10`, `torch==2.4.0`, `numba==0.59.1`, and the NumPy-1-era MIDAS backends (`calibrate-v2 0.3.3`, `integrate-v2 0.1.0`, `calibrate 0.2.3`, `hkls 0.4.1`, `distortion 0.2.0`) via pip; then `-e .`. Dropped the conda pytorch/cpuonly lines + pytorch channel (torch now via pip). `pyproject.toml`: lowered `midas-calibrate` floor 0.2.7 → 0.2.3 so the proven base version satisfies `-e .`. README gains a "Recreating the environment" section (remove + create), a conda-solver/libarchive note, and the CPU-only torch install note. Verified end-to-end: fresh env resolves clean; GUI

- a real nickel-frame integration run (numpy 1.26.4 + torch 2.4.0). **Files:** `environment.yml`, `pyproject.toml`, `README.md`. **Roll back:** `git revert 68313f8` (returns to ranged deps; a fresh env would again risk the NumPy-2/torch mismatch). To move forward to a NumPy-2 stack instead, bump `torch>=2.3` and the backends to their numba-0.66 releases together.

## 3a6ecb9 — Calibrate Results: all-text multi-column readout (drop distortion table) (2026-07-14)

**Effect:** The Calibrate **Results** tab is now entirely text. Removed the distortion `QTableWidget`; the distortion coefficients appear in the same multi-column parameter grid, with the paramtest `p0-p14` slots relabelled to their names (`iso_R2`, `a1`, `phi1`, ...) via `helpers._PARAMSTEST_DISTORTION`. Larger font (12 pt) and roomier row/column spacing for readability. Removed the `DistortionTable` import + `set_distortion` call. **Files:** `tab_calibrate.py`, `documentation/gui_documentation.md`. **Roll back:** `git revert 3a6ecb9` (restores the named-distortion table widget below the grid). `DistortionTable` still exists in `widgets.py` (used only here), so a revert is clean.

## b1642fa — Fix calibration TypeError (initial\_BC\_y) + add scikit-image (2026-07-14)

**Effect:** Fixes calibration failing with `calibrate()` got an unexpected keyword argument `'initial_BC_y'` on the pinned (base-matching) `midas-calibrate-v2 0.3.3`, whose `calibrate()` has no beam-centre seed parameter (only newer backends add it). New `calib._supported_kwargs(fn, kwargs)` filters kwargs to the installed callable's signature (keeps `initial_Lsd`, drops the unsupported `initial_BC_y/z`, logs it), so the GUI tolerates backend-version signature drift. Also pins **scikit-image=0.23.2** in `environment.yml` — `midas-calibrate-v2`'s `auto_seed_calibrant`

needs it; without it, seeding degrades to the coarse arc fallback. Verified: manual-seed one\_shot calibration of the shipped ceria runs end-to-end (numpy 1.26.4 + torch 2.4.0). **Files:** `calib.py`, `environment.yml`. **Roll back:** `git revert b1642fa` (reintroduces the `TypeError` on backends lacking the BC-seed args, and removes the scikit-image pin).

## **df544d2 — Data Viewer: tilt/distortion-aware radial integration (2026-07-14)**

**Effect:** When a loaded calibration file carries the full geometry (tilts + distortion), the Data Viewer's radial integration goes through the MIDAS engine ( `build_integration_context` + `integrate_frame` ) instead of concentric-circle binning, so pixels map through the calibrated tilts/distortion. `_load_calibration` now also reads the full geometry via `geometry_fields_from_file` (falls back to scalar/circle mode if absent) and shows the active mode. `_midas_radial` caches the binning geometry per (geometry, R-bin, image shape, static mask) and reuses it across frames (fast `hard` kernel; loader's static composite mask so the cache holds); `_radial_integrate` branches to it with a guarded fallback to circle binning on error. **Files:** `tab_view.py`, `documentation/gui_documentation.md`. **Roll back:** `git revert df544d2` (reverts to circle-only radial integration in the Data Viewer). Self-contained; the MIDAS-engine primitives it reuses are unchanged.

## **bc86d74 — Data Viewer: Top-N pixels + mask-aware stats + full-geometry receive (2026-07-19)**

**Effect:** Adds a "Top-N pixels" toolbar toggle that marks the N brightest pixels (crosshair + circle) and feeds their values to the intensity-statistics panel, re-ranking live on frame/mask changes. Masked pixels (mask file + intensity-range mask) are excluded from the Top-N ranking and its stats/histogram. `_midas_radial` now unions the static mask with the per-frame intensity-range mask and fingerprints the mask by content so the cached binning context rebuilds when the mask changes. The tab also accepts a *full* geometry dict (tilts + distortion + detector size) from Calibrate and routes integration through the MIDAS engine when it arrives. The stats panel readout box auto-sizes to its content (no inner scrollbar) with the histogram as the single flexible child; `DataLoaderPanel` is now a `QWidget` hosting a scroll area + stats panel in a draggable vertical splitter. **Files:** `tab_view.py`, `widgets.py`, `documentation/gui_documentation.md`. **Roll back:** `git revert bc86d74`.

## **b0a5cc3 — Calibrate: per-coefficient distortion, frame averaging, seed feedback (2026-07-19)**

**Effect:** Adds `DistortionRefineDialog` (behind the "..." next to the Distortion checkbox) to pick any of the 15 harmonics or use  $\eta$ -fold presets; `calib.py` maps each selected v2 harmonic to its v1 p-

slot (via `DISTORTION_ISO / DISTORTION_PRESETS` in constants). Adds an "Average frames" card (start<sup>END</sup> skip, streamed via `DataLoaderPanel.average_frames`), seed tilts (tx/ty/tz) carried into the v1 params, a "Feed result back to seed" option, and a "→ Send to Data Viewer" that now sends the full geometry (tilts + distortion + detector size). Drops the 310px bottom-panel cap and adds a non-collapsible splitter. **Files:** `calib.py`, `tab_calibrate.py`, `dialogs.py`, `constants.py`, `widgets.py`, `documentation/gui_documentation.md`. **Roll back:** `git revert b0a5cc3`.

### **b6f1681 — Batch: read-only "Calibration values" card (2026-07-19)**

**Effect:** Adds a Calibration values card to the Batch parameter column showing the geometry actually driving integration ( $\lambda$ , Lsd, BC, tilts, pixel sizes, detector size, non-zero distortion count), resolved from the live Tab-2 result or a calibration file and refreshed on source/path change. Also lets the Log panel fill its splitter space. **Files:** `tab_batch.py`, `documentation/gui_documentation.md`. **Roll back:** `git revert b6f1681`.

### **49623ae — Pump Probe: publication plotting, delay colour-bar, default mask (2026-07-19)**

**Effect:** Publication-quality plot defaults (clean 2.0px lines, 13pt fonts, dashed grid, refactored `_rainbow_cmap`); a continuous delay→colour bar ( `GradientLegend` ) replaces the many-row legend past a threshold, with left/right legend anchoring; a log-delay x-axis option for kinetics; and a default detector mask wired via new `DataLoaderPanel.add_mask_file / MaskSelector.add_file_source`. TR-XRD defaults now point at a local-only `test_data_pump_probe/` folder kept out of git. **Files:** `tab_pumpprobe.py`, `constants.py`, `widgets.py`, `documentation/gui_documentation.md`. **Roll back:** `git revert 49623ae`.

### **b251ca8 — chore: gitignore context, internal docs, large sample sets (2026-07-19)**

**Effect:** Ignores the local `.context/` scaffold and `CLAUDE.md`, the internal-only PDF calculation write-up, and the large local sample dirs ( `test_data/17BM/`, `test_data/test_s25ide/` ) via directory-level ignores that override the `test_data/**` re-includes. **Files:** `.gitignore`. **Roll back:** `git revert b251ca8`.

### **1769f69 — Data Viewer: live PV stream (Live Data card) (2026-07-20)**

**Effect:** Subscribes to an EPICS PVA image PV via `pvapy` and pushes frames straight through the existing image/corrections/ring/radial-integration pipeline — no separate viewer window. "Live Data" is a collapsible card (own title-bar checkbox) above the Data card; unchecking stops an active



stream. Data card gets a small ↻ reload button to restore the static file/folder/HDF5 source after stopping a stream. `pvapy==5.4.1` is now a required, pinned dependency (chosen over latest to avoid upgrading the numpy/pyqtgraph pins). `MainWindow` calls a generic `tab.shutdown()` hook on close so a live stream stops cleanly on app exit. **Files:** `widgets.py`, `tab_view.py`, `app.py`, `pyproject.toml`, `environment.yml`, `documentation/gui_documentation.md`, `tests/test_live_stream.py`. **Roll back:** `git revert 1769f69`.

## **0b4326d — Fix: cap native thread pools to prevent calibration crash (2026-07-22)**

**Effect:** Clicking "Run Calibration" crashed the app outright — a native, uncatchable `Bus error`, not a Python exception. Root cause: `CalibrationWorker` (a `QThread`) triggers a multi-threaded `numpy.linalg.inv()` deep inside `midas-calibrate-v2`'s HKL/ring generation (likely surfaced by the recent 0.3.3 → 0.5.2 bump); running from a `QThread` while the Qt event loop is alive, that races against PyTorch's own OpenMP pool and corrupts memory. Reproduced deterministically at two call sites (`numpy.linalg.inv` during ring generation, `scipy.ndimage.median_filter` during seeding); both disappear once native thread pools are capped to 1. Every heavy operation in this app runs inside a `QThread` (17 worker classes in `workers.py`), so the fix caps `OPENBLAS_NUM_THREADS` / `OMP_NUM_THREADS` / `MKL_NUM_THREADS` / `VECLIB_MAXIMUM_THREADS` / `NUMEXPR_NUM_THREADS` to 1 process-wide via `os.environ.setdefault` in `_paths.py` — already home to the sibling `KMP_DUPLICATE_LIB_OK` workaround for the same underlying duplicate-OpenMP-runtime issue — rather than patching each worker individually. Verified: offscreen run of the full calibration pipeline to convergence with no crash; `pytest` 15/15 green. **Files:** `midas_gui/_paths.py`. **Roll back:** `git revert 0b4326d` (restores multi-threaded BLAS; calibration/other workers become exposed to this native-crash class again on macOS).

## **234ded9 — Live viewer: preserve manual colormap/level changes across frames (2026-07-24)**

**Effect:** `ImageViewer._redisplay` recomputed `vmin%/vmax%` percentile levels on every incoming frame (live or otherwise), silently overwriting a level range the user had dragged on the pyqtgraph histogram mid-acquisition. Now tracks manual histogram edits via `sigLevelsChanged` (guarded against feedback from our own programmatic `setImage` calls with a suspend flag) and keeps pinning to the last manual levels until the user explicitly changes the `vmin%/vmax%` spin boxes, which resets back to auto. **Files:** `widgets.py`. **Roll back:** `git revert 234ded9`.

## **69f470c — Data Viewer: widen the image/radial-plot splitter handle (2026-07-24)**

**Effect:** The vertical splitter between the image view and the radial-integration panel had no explicit handle width (unlike the outer horizontal splitter), making it thin and hard to grab. Set `setHandleWidth(8)` to match. **Files:** `tab_view.py`. **Roll back:** `git revert 69f470c`.

## 96f8add — Radial profile: stop forcing Y-axis min to -1000 on every update (2026-07-24)

**Effect:** `ProfileViewer._replot` unconditionally called `setYRange()` with a hardcoded -1000 floor on every redraw, clobbering any manual Y-range the user set. The auto lower bound is now `0.9 * current data minimum`; once the user manually changes the Y range (drag, wheel-zoom, or the axis context menu's numeric entry — detected via `sigYRangeChanged` with the same suspend-flag pattern as `234ded9`), that value is kept for the rest of the session instead of being recomputed. Also dropped the hardcoded `setLimits(yMin=-1000/1)` hard floors. **Files:** `widgets.py`. **Roll back:** `git revert 96f8add`.

## 3eb4a44 — Calibrate: recolor Pick-Ring fit overlay to blue (2026-07-24)

**Effect:** `PickableImageViewer`'s Pick-Ring points, fitted circle, and fitted-center marker used the same amber ( `#f0c060` ) as the Data Viewer's Simulate-rings overlay, making the two indistinguishable when both were visible on the same image. Pick-Ring's fit overlay is now blue ( `#2a7fd4` ); the separate Pick-BC "+" click marker (already `#00aaff` ) and Simulate-rings amber were left unchanged. **Files:** `widgets.py`. **Roll back:** `git revert 3eb4a44`.

## ecf6d01 — Data Viewer: make Simulate rings a live toggle instead of one-shot (2026-07-24)

**Effect:** "Simulate rings" is now a checkable toggle rather than a single-click action. While on, it resimulates automatically whenever material, lattice constants, space group, or geometry (wavelength/Lsd/pixel size/max 2 $\theta$ ) change — mirroring the existing `_rad_auto` -style "recompute on change" idiom already used for radial integration. Beam-centre edits still just reposition the existing rings (unchanged; ring radii don't depend on BC), so no wiring was added there. **Files:** `tab_view.py`. **Roll back:** `git revert ecf6d01`.

## aa82cb6 — Preferences: add Devices tab; Data Viewer Live PV becomes a device dropdown (2026-07-24)

**Effect:** New **Preferences ▶ Devices** tab (add/remove/edit rows of name + prefix + PVA suffix), following the existing Materials/Calibrants table-tab pattern; persisted as a new `"devices"` list in the JSON config (replaces wholesale when present, same as materials/calibrants). Ships pre-filled with the 20-ID-D detectors extracted from the beamline's `make_det(...)` blueprint — `oryx20idd` ( `20idd0R1:` ), `s20idPil` ( `20idPil` ), `pg4` ( `1idPG4:` ), `gh1s` ( `20idGH1s:` ), `s20varex1` ( `20IDFF:` ) — all with PVA suffix `Pva1:Image`. The Data Viewer's Live Data "Live PV" field is now an editable combo box ( `_NoScrollComboBox` ) populated from this list: picking a device by name fills in its full

PV ( `prefix + PVA suffix` ); typing any other PV by hand still works unchanged, with the same example placeholder text. **Files:** `constants.py` , `prefs_dialog.py` , `widgets.py` , `tests/test_live_stream.py` . **Roll back:** `git revert aa82cb6` (Devices tab and the Live PV dropdown both go; the Live PV field reverts to a plain text box).

## c156c62 — Fix image-view autorange drift; bound pan/zoom to image (2026-07-24)

**Effect:** Two bugs in the shared `ImageViewer` (Data Viewer / Mask Builder / Calibrate all use it): (1) the crosshair `InfiniteLine`s were added to the pyqtgraph `ViewBox` without `ignoreBounds=True` , so their position — which follows the mouse on every hover event — fed the auto-range bounds calculation. Once the bottom-left **A** (auto-range) button enabled *continuous* auto-range, the view kept re-fitting itself to wherever the cursor was, and only a right-click ▶ View All (which disables continuous auto-range as a side effect) made it static again. Fixed by excluding the crosshairs from bounds via `ignoreBounds=True` . (2) Pan/zoom had no limits, so scroll/drag could wander arbitrarily far from the image or zoom out indefinitely into empty space. Added `ViewBox.setLimits()` (computed from the image shape in `set_image()` ): pan is bounded to roughly half an image-width/height of margin past each edge, and min/max zoom range are tied to the image size. **Files:** `widgets.py` ( `ImageViewer.set_image` , new `_apply_view_limits` ). **Roll back:** `git revert c156c62` (crosshair can drift the view again under continuous auto-range; pan/zoom become unbounded again).

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## Rollback recipes (common intents)

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- **Undo the Pump Probe tab only:** `git revert 590b410` removes Pump Probe *and* modular tabs. To drop Pump Probe alone, hand-remove `midas_gui/tab_pumpprobe.py` , its import/construction/wiring in `app.py` , `PumpProbeWorker` in `workers.py` , and its entry in `constants.OPTIONAL_TABS / DEFAULT_VISIBLE_TABS` .
- **Undo modular tabs (show all 10 fixed again):** remove the `apply_tab_visibility` logic in `app.py` (add all tabs directly), the Tabs section in `prefs_dialog.py` , and the tab-list constants — see the `590b410` diff for the exact hunks.
- **Undo the config system:** `git revert d30be2c` — but first revert `590b410` 's config touches, since it extends the same `ui` overlay and Preferences dialog.
- **Revert the azimuthal-averaging change:** don't blanket-revert `41f28f5` (bundled with UI); instead set the `weighted` default back to the legacy  $\eta$ -bin mean in the batch/pump workers.
- **Go back to the pre-3-panel single-column tabs:** `aa9395e` is the pivot; a revert is large and cascades to later tabs — prefer fixing forward.

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*Generated from `git log` . To refresh after new commits: `git log --reverse --stat --date=short` and append entries above.*